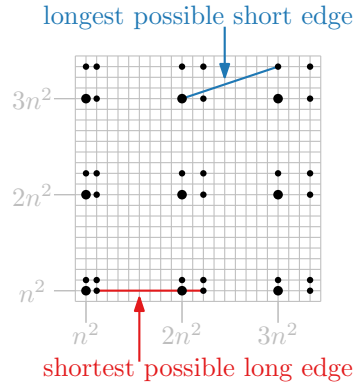


respective circles intersect on ℓ_{k+1} . Observe that with this operation, the minimum distance between any two vertices is still d .

Next, we process first pairs. Observe that no vertex can be involved in several first pairs. Otherwise it is not on the outer face. However, a vertex can be contained in a first and another pair and it is also possible that a vertex closing another face is contained in a first pair. So some of the vertices might already have a fixed position. However, it is not possible that both vertices of a first pair are contained in another pair or close a face. Otherwise, the predecessor of the first pair was an internal vertex. So, we first process the first pairs that share a vertex with another pair or where one of its vertices close a face and reduce their distance to d . Then we can also process all the other first pairs. In the end all pairs have distance d and d is still the minimum distance between any two vertices on ℓ_{k+1} . This concludes the construction for Items 1 and 2.

The strips are defined as follows: Let $u \in V_{k-1}$ and let $v_1, \dots, v_h$ be the children of u from left to right. We partition the strip S_u into the strips $S_{v_1}, \dots, S_{v_h}$ by splitting S_u at an arbitrary spot between $x_{v_{i-1}}^r$ and $x_{v_i}^l$, for $i = 2, \dots, h$. If a vertex has two parents then its strip is the union of the two portions obtained from its parents.

By construction, all edges of G_s have length at most one. It remains to show that edges in E_ℓ have length greater than one. So let $w_j w_k \in E_\ell$ with $w_j \in V_j$, $w_k \in V_k$, and $j \leq k$. Since G is bipartite, we can exclude $k = j$ and $k = j + 2$. If $k - j \geq 3$ then $y_k - y_j > k - 1 - j \geq 2$. Hence, the distance between w_j and w_k is greater than one. If $k - j = 1$ then the parent w of w_k is in V_j . By construction, we know that the distance between w_j and any child of w is greater than one. $\blacktriangleleft$



■ **Figure 8** For each grid point (i, j) , $1 \leq i \leq n$, $1 \leq j \leq n$ there are four possible points. If $i > 1$, the x-coordinate is in^2 if the edge between $(i-1, j)$ and (i, j) is short and $in^2 + i$ otherwise. If $j > 1$, the y-coordinate is jn^2 if the edge between $(i, j-1)$ and (i, j) is short and $jn^2 + j$ otherwise.

► **Theorem 9.** A dichotomous ordinal graph $G = (V, E_s \cup E_\ell)$ admits a geometric realization if the set of short edges induces a subgraph of the grid.

Proof. Extend $G_s = (V, E_s)$ by the remaining grid edges and require the new edges to be long. We now perturb the grid. For each grid point (i, j) , $1 \leq i \leq n$, $1 \leq j \leq n$ of the original grid, there are four possible choices with the x-coordinates in^2 or $in^2 + i$ and the y-coordinates jn^2 or $jn^2 + j$. See Fig. 8. In the case where $i = 1$ choose as x-coordinate n^2 or $n^2 + 1$. Similarly, if $j = 1$ choose as y-coordinate n^2 or $n^2 + 1$. If $i > 1$, we choose the x-coordinate in^2 if the edge between $(i-1, j)$ and (i, j) is short and $in^2 + i$ otherwise.